

Examen la logică computațională
Biletul nr. 2

1. (a) $101011,111_{(2)} = ?_{(10)}$

$$N = 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 + 1 \cdot 2^{-1} + 1 \cdot 2^{-2} + 1 \cdot 2^{-3} =$$

$$= 32 + 8 + 2 + 1 + 0,5 + 0,25 + 0,125 = 43,875$$

$$\Rightarrow 101011,111_{(2)} = 43,875_{(10)}$$

$$43,875_{(10)} = ?_{(2)}$$

$$\begin{array}{l} 43 = 2 \cdot 21 + 1 \\ 21 = 2 \cdot 10 + 1 \\ 10 = 2 \cdot 5 + 0 \\ 5 = 2 \cdot 2 + 1 \\ 2 = 2 \cdot 1 + 0 \\ 1 = 2 \cdot 0 + 1 \end{array} \uparrow$$

$$0,875 \cdot 2 = 1 + 0,75 \rightarrow a_{-1} = 1$$

$$0,75 \cdot 2 = 1 + 0,5 \rightarrow a_{-2} = 1$$

$$0,5 \cdot 2 = 1 \rightarrow a_{-3} = 1$$

$$43_{(10)} = 101011_{(2)}$$

$$0,875_{(10)} = 0,111_{(2)}$$

$$\Rightarrow 43,875_{(10)} = 101011,111_{(2)}$$

(b) • reprezentare prin mărime și semn (cod direct):

$$[+26]_{\text{dir}} = 0 \ 11010; \quad [-26]_{\text{dir}} = 1 \ 11010$$

• reprezentare prin complement față de 2 (cod complementar):

$$[+26]_{\text{comp}} = 0 \ 11010;$$

$$1 \ 00101 +$$

$$\underline{ 1}$$

$$1 \ 00110$$

$$\Rightarrow [-26]_{\text{comp}} = 1 \ 00110$$

$$26 = 2 \cdot 13 + 0 \uparrow$$

$$13 = 2 \cdot 6 + 1$$

$$6 = 2 \cdot 3 + 0$$

$$3 = 2 \cdot 1 + 1$$

$$1 = 2 \cdot 0 + 1$$

$$26_{(10)} = 11010$$

• reprezentare prin complement față de 1 (cod invers):

$$[+26]_{\text{inv}} = 0 \ 11010; \quad [-26]_{\text{inv}} = 1 \ 00101$$

$$2. \bar{x}_1 x_3 + x_2 (\bar{x}_1 + x_3) = \bar{x}_1 x_3 + \bar{x}_1 x_2 + x_2 x_3$$

$$\text{Notăm } f_1(x_1, x_2, x_3) = \bar{x}_1 x_3 + x_2 (\bar{x}_1 + x_3)$$

$$f_2(x_1, x_2, x_3) = \bar{x}_1 x_3 + \bar{x}_1 x_2 + x_2 x_3$$

x_1	x_2	x_3	$\bar{x}_1 x_3$	$\bar{x}_1 + x_3$	$x_2 (\bar{x}_1 + x_3)$	$f_1(x_1, x_2, x_3)$	$\bar{x}_1 x_2$	$x_2 x_3$	$f_2(x_1, x_2, x_3)$
0	0	0	0	1	0	0	0	0	0
0	0	1	1	1	0	1	0	0	1
0	1	0	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0	0
1	0	1	0	1	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
1	1	1	0	1	1	1	0	1	1

⇒ Valorile din funcțiile $f_1(x_1, x_2, x_3)$ și $f_2(x_1, x_2, x_3)$ sunt identice pentru toate combinațiile valorilor argumentelor x_1, x_2, x_3 ⇒

$$\rightarrow f_1(x_1, x_2, x_3) = f_2(x_1, x_2, x_3) \Leftrightarrow \bar{x}_1 x_3 + x_2 (\bar{x}_1 + x_3) = \bar{x}_1 x_3 + \bar{x}_1 x_2 + x_2 x_3$$

$$3. \overline{(x_1 + x_2 \bar{x}_3)(\bar{x}_1 + \bar{x}_3 + x_2)} = \overline{(x_1 + x_2 \bar{x}_3)(x_1 x_3 + x_2)} = \overline{(x_1 + x_2 \bar{x}_3)} + \overline{(x_1 x_3 + x_2)} =$$

$$= \bar{x}_1 (\bar{x}_2 + x_3) + (\bar{x}_1 + \bar{x}_3) \bar{x}_2 = \bar{x}_1 \bar{x}_2 + \bar{x}_1 x_3 + \bar{x}_1 \bar{x}_2 + \bar{x}_2 \bar{x}_3 = \bar{x}_1 \bar{x}_2 + \bar{x}_1 x_3 + \bar{x}_2 \bar{x}_3 =$$

$$= \bar{x}_1 (\bar{x}_2 + x_3) + \bar{x}_2 \bar{x}_3$$

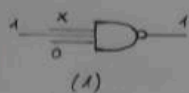
$$4. f(x_1, x_2, x_3) = x_2 + x_1 \bar{x}_3 + \bar{x}_1 \bar{x}_2$$

$$f^{FCB}(x_1, x_2, x_3) = (x_1 + \bar{x}_1) x_2 (x_3 + \bar{x}_3) + x_1 (x_2 + \bar{x}_2) \bar{x}_3 + \bar{x}_1 x_2 (x_3 + \bar{x}_3)$$

$$f^{FCB}(x_1, x_2, x_3) = x_1 x_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + \bar{x}_1 \bar{x}_2 x_3 + \bar{x}_1 \bar{x}_2 \bar{x}_3$$

$$f^{FCB}(x_1, x_2, x_3) = x_1 x_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3$$

5. (a)

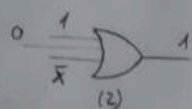


Poartă logică SI-NU cu 3 intrări ($x, 1, 0$) și o ieșire (1).

Notare $f_1(a, b, c) = \overline{x} \cdot 1 \cdot 0 = 1$

a	b	c	$f_1(a, b, c)$
0	0	0	1
0	0	1	1
* 0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
* 1	1	0	1
1	1	1	0

Ptr. combinațiile valorilor argumentelor: (0, 1, 0) și (1, 1, 0), $f_1(a, b, c) = 1 \Rightarrow x = 0$ sau $x = 1$



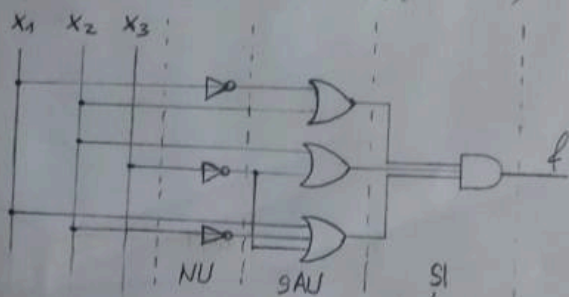
Poartă logică SAU cu 3 intrări ($1, 0, \overline{x}$) și o ieșire (1).

Notare $f_2(a, b, c) = 1 + 0 + \overline{x} = 1$

a	b	c	$f_2(a, b, c)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
* 1	0	0	1
* 1	0	1	1
1	1	0	1
1	1	1	1

Pentru combinațiile valorilor argumentelor: (1, 0, 0) și (1, 0, 1), $f_2(a, b, c) = 1 \Rightarrow \overline{x} = 0$ sau $\overline{x} = 1 \Rightarrow x = 1$ sau $x = 0$

(b) $f^{FMC}(x_1, x_2, x_3) = (\overline{x}_1 + x_2)(x_2 + \overline{x}_3)(x_1 + \overline{x}_2 + \overline{x}_3)$



6. (a) $f^{FCD}(x_1, x_2, x_3, x_4) = \sum(1, 3, 4, 5, 6, 9, 11, 12, 13, 14)$

I. Metoda diagrama Karnaugh

	x_3x_4	00	01	11	10
00		0	1	1	0
01		1	1	1	1
11		1	0	0	1
10		0	1	1	0

$x_2\bar{x}_3$
 $\bar{x}_3x_4$
 $\bar{x}_2x_4$
 $x_2\bar{x}_5$

	x_1x_2	00	01	11	10	
x_3x_4	00	0	1	1	0	$\bar{x}_3\bar{x}_4$
	01	1	1	1	1	
	11	1	0	0	1	$x_3x_3x_4x_4$
	10	0	1	1	0	

$\Rightarrow f(x_1, x_2, x_3, x_4) = x_2 \bar{x}_1 + \bar{x}_2 x_4 + \bar{x}_3 x_1 + x_2 \bar{x}_3$
 $f^{FCD}(x_1, x_2, x_3, x_4) = x_2 \bar{x}_1 + \bar{x}_2 x_4 + \left\{ \begin{matrix} \bar{x}_3 x_1 \\ x_2 \bar{x}_3 \end{matrix} \right\}$

$\Rightarrow \bar{f}^{FCD}(x_1, x_2, x_3, x_4) = \bar{x}_2 \bar{x}_4 + x_2 x_3 x_4$
 $\Rightarrow \bar{f}^{FCD}(x_1, x_2, x_3, x_4) = \dots$
 $= f^{FMC}(x_1, x_2, x_3, x_4) = (x_2 + x_4)(\bar{x}_2 + \bar{x}_3 + \bar{x}_4)$

II. Metoda Quine-McCluskey

Etapas I:

Subcuburi 0-dimensionale						Subcuburi 1-dimensionale						Subcuburi 2-dimensionale					
Grupa	Indici	x_1	x_2	x_3	x_4	Grupa	Indici	x_1	x_2	x_3	x_4	Grupa	Indici	x_1	x_2	x_3	x_4
1	1	0	0	0	1 ✓	1	1,3	0	0	-	1 ✓	1	1,3,9,11	-	0	-	1 (a)
	4	0	1	0	0 ✓		1,5	0	-	0	1 ✓		1,5,9,13	-	-	0	1 (b)
2	3	0	0	1	1 ✓		1,9	-	0	0	1 ✓		1,9,3,11	-	0	-	1
	5	0	1	0	1 ✓		4,5	0	1	0	- ✓		1,9,5,13	-	-	0	1
	6	0	1	1	0 ✓		4,6	0	1	-	0 ✓		4,5,12,13	-	1	0	- (c)
	9	1	0	0	1 ✓	4,12	-	1	0	0 ✓	4,6,12,14	-	1	-	0 (d)		
3	12	1	1	0	0 ✓	2	3,11	-	0	1	1 ✓	4,12,5,13	-	1	0	-	
	11	1	0	1	1 ✓		5,13	-	1	0	1 ✓	4,12,6,14	-	1	-	0	
	13	1	1	0	1 ✓		6,14	-	1	1	0 ✓						
	14	1	1	1	0 ✓		9,11	1	0	-	1 ✓						
							9,13	1	-	0	1 ✓						
							12,13	1	1	0	- ✓						
							12,14	1	1	-	0 ✓						

$\Rightarrow f(x_1, x_2, x_3, x_4) = \bar{x}_2 \bar{x}_4 + \bar{x}_3 x_1 + x_2 \bar{x}_3 + x_2 \bar{x}_4$

Etapa II

Implicanți primi + indici	Termeni canonici									
	1	3	4	5	6	9	11	12	13	14
(a) 1, 3, 9, 11	*	*				*	*			
(b) 1, 5, 9, 13	*			*		*			*	
(c) 4, 5, 12, 13			*	*				*	*	
(d) 4, 6, 12, 14			*		*			*		*

$$\text{IPE} \rightarrow \bar{x}_2 x_4$$

$$\text{IPE} \rightarrow x_2 \bar{x}_4$$

Cei 2 IPE acoperă termenii canonici: 1, 3, 4, 6, 9, 11, 12, 14, $\Rightarrow$

$\rightarrow$ Mai rămân de acoperit termenii canonici: 5, 13 (continuiți de căutare dintre implicanții primi b și c)

$$\Rightarrow f^{FMD}(x_1, x_2, x_3, x_4) = \bar{x}_2 x_4 + x_2 \bar{x}_4 + \begin{cases} \bar{x}_3 x_4 \\ x_2 \bar{x}_3 \end{cases}$$

$$(b) \begin{cases} f^{FCD}(x_1, x_2, x_3, x_4) = \Sigma(3, 4, 6, 7, 9, 11, 12) \\ d = 1 = 15 = * \end{cases}$$

Metoda diagramelor Karnaugh

$x_1 x_2$	00	01	11	10
$x_3 x_4$	0	1	1	0
01	d	0	0	1
11	1	1	d	1
10	0	1	0	0

$$\bar{x}_2 \bar{x}_3 \bar{x}_4$$

$x_1 x_2$	00	01	11	10
$x_3 x_4$	0	1	1	0
01	d	0	0	1
11	1	1	d	1
10	0	1	0	0

$$\Rightarrow f^{FMD}(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_2 \bar{x}_4 + x_2 \bar{x}_3 \bar{x}_4 + \bar{x}_2 x_4 + x_3 x_4$$

$$\Rightarrow f^{FMD}(x_1, x_2, x_3, x_4) = \bar{x}_2 \bar{x}_4 + \begin{cases} x_2 \bar{x}_3 \bar{x}_4 \\ \bar{x}_1 \bar{x}_3 x_4 \end{cases} + \begin{cases} x_1 x_2 x_3 \\ x_1 x_2 x_4 \end{cases}$$

$$\Rightarrow f^{FMD}(x_1, x_2, x_3, x_4) = (x_2 + x_4) \cdot (\bar{x}_2 + x_3 + \bar{x}_4) \cdot (x_1 + x_2 + x_3) \cdot (x_1 + x_2 + \bar{x}_4)$$